

# Lagrangean Method II & Indirect Utility

Alejandro Melo Ponce



Stony Brook University

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# The general constrained optimization problem

- Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $g : \mathbb{R}^n \rightarrow \mathbb{R}$  and  $h : \mathbb{R}^n \rightarrow \mathbb{R}$  be multivariate real-valued functions.
- The standard constrained optimization problem is given by:

$$\begin{aligned} \max_{(x_1, \dots, x_n) \in \mathbb{R}^n} : & \quad f(x_1, \dots, x_n) \\ \text{such that} & \quad g(x_1, \dots, x_n) = M \\ & \quad h(x_1, \dots, x_n) \leq Q \end{aligned} \tag{P}$$

- Where  $M, Q$  are two real numbers.
- We have two constraints, one that should hold with equality, and one that can either hold with equality or strict inequality at an optimum.
- We can solve this problem with the method of Lagrange.

# Lagrangian Example: A utility maximization problem with equality constraints

- Consider the following problem:

$$\max_{(x,y) \in \mathbb{R}^2} : \sqrt{x} + \sqrt{y} \quad \text{subject to } p_x x + p_y y \leq I$$

- For this utility function, the constraint binds with **equality** because the utility is monotone, i.e. the more the better.
- The Lagrangian is:

$$L(x, y, \lambda) = \sqrt{x} + \sqrt{y} + \lambda(I - p_x x - p_y y)$$

- FOC's are:

$$\frac{\partial L}{\partial x} = \frac{1}{2\sqrt{x^*}} - \lambda^* p_x = 0$$

$$\frac{\partial L}{\partial y} = \frac{1}{2\sqrt{y^*}} - \lambda^* p_y = 0$$

$$\frac{\partial L}{\partial \lambda} = I - p_x x^* - p_y y^* = 0$$

## Example continued

- The easiest thing to do here is to first get rid of  $\lambda^*$ : using the first two equations we get

$$\lambda = \frac{1}{2\sqrt{x^*}p_x} \quad \lambda = \frac{1}{2\sqrt{y^*}p_y}$$

- Combining them we get:  $2\sqrt{x^*}p_x = 2\sqrt{y^*}p_y \Rightarrow x^*p_x^2 = y^*p_y^2$ .
- Using the previous result and the restriction  $I - p_x x^* - p_y y^* = 0$  yields

$$p_x x^* + p_y \left( x^* \frac{p_x^2}{p_y^2} \right) = I \rightarrow x^* \left( p_x + \frac{p_x^2}{p_y} \right) = I$$
$$\frac{p_x p_y + p_x^2}{p_y} x^* = I \rightarrow x^* = \frac{I p_y}{p_x p_y + p_x^2}, \quad y^* = \frac{I p_x}{p_y^2 + p_x p_y}$$

# The Method of Lagrange multipliers: Inequality Constraints

- Now consider the version of problem (P) in which we only have the inequality constraint.

$$\begin{aligned} \max_{(x_1, \dots, x_n) \in \mathbb{R}^n} : & \quad f(x_1, \dots, x_n) \\ \text{such that} & \quad h(x_1, \dots, x_n) \leq Q \end{aligned} \quad (\text{PE})$$

- We define the **Lagrangian function** as before

$$L(x_1, \dots, x_n, \lambda) = f(x_1, \dots, x_n) + \lambda(Q - h(x_1, \dots, x_n)) \quad (1)$$

## Solving the problem

- We are going to take first order conditions of the Lagrangian with respect to  $x_1, \dots, x_n$

$$[x_1]: f_1(x_1, \dots, x_n) - \lambda h_1(x_1, \dots, x_n) = 0$$

$$[x_2]: f_2(x_1, \dots, x_n) - \lambda h_2(x_1, \dots, x_n) = 0$$

$$\vdots$$

$$[x_n]: f_n(x_1, \dots, x_n) - \lambda h_n(x_1, \dots, x_n) = 0$$

But now the condition with respect to  $\lambda$  changes

$$[\lambda]: Q - h(x_1, \dots, x_n) \geq 0$$

Furthermore, we need to add what is called **Complementary Slackness conditions**

$$\lambda \geq 0, \quad \lambda[Q - h(x_1, \dots, x_n)] = 0$$

## Example: Utility maximization when the constraint doesn't bind

- Consider the following problem:

$$\max_{(x,y) \in \mathbb{R}^2} : 100 - (x-1)^2 - (y-1)^2 \quad \text{subject to } p_x x + p_y y \leq I$$

- This is a utility function representing the preferences from HW 1 Ex. 6. This Utility function has a bliss point at  $(1, 1)$
- The Lagrangian is:

$$L(x, y, \lambda) = 100 - (x-1)^2 - (y-1)^2 + \lambda(I - p_x x - p_y y)$$

- FOC's are:

$$\frac{\partial L}{\partial x} = -2(x^* - 1) - \lambda^* p_x = 0$$

$$\frac{\partial L}{\partial y} = -2(y^* - 1) - \lambda^* p_y = 0$$

$$\frac{\partial L}{\partial \lambda} = I - p_x x^* - p_y y^* \geq 0,$$

$$\lambda^* \geq 0, \quad \lambda^* [I - p_x x^* - p_y y^*] = 0$$

## Example continued

- We need to consider two cases,  $\lambda^* = 0$  and  $\lambda^* > 0$ .
- First case:  $\lambda^* = 0$  or the **constraint is not binding**.
  - Then  $I - p_x x^* - p_y y^* > 0$  to satisfy complementary slackness.
  - From the first order conditions on  $x$  and  $y$  we immediately get that since  $\lambda^* = 0$ ,  $x^* = y^* = 1$
- Second case:  $\lambda^* > 0$  or the **constraint is binding**.
- Then  $I - p_x x^* - p_y y^* = 0$  to satisfy complementary slackness.
- From the first order conditions on  $x$  and  $y$  we get that

$$x^* - 1 = -\lambda^* p_x$$

$$y^* - 1 = -\lambda^* p_y$$

Thus, since  $\lambda^* > 0$  we necessarily need

$$x^* < 1, y^* < 1, \quad \text{Furthermore}$$

$$\frac{x^* - 1}{y^* - 1} = \frac{p_x}{p_y}, \quad \text{this is } MRS = p_x/p_y$$

## Example continued

- From  $\frac{x^*-1}{y^*-1} = \frac{p_x}{p_y}$  we get  $x^* = \frac{p_x}{p_y} + 1$
- Substituting into the Budget constraint

$$p_x \left( \frac{p_x}{p_y} (y^* - 1) + 1 \right) + p_y y^* = I,$$

- Which after many simplifications, yields

$$y^* = \frac{Ip_y + p_x^2 - p_x p_y}{p_x^2 + p_y^2}$$

- Similarly

$$x^* = \frac{Ip_x + p_y^2 - p_x p_y}{p_x^2 + p_y^2}$$

# Indirect utility

- Let's go again to the utility maximization problem

$$\max_{(x,y) \in \mathbb{R}^2} : U(x,y) \quad \text{subject to } p_x x + p_y y \leq I$$

- Recall that solving this problem, yields the **Marshallian demands** of  $x$  and  $y$ , i.e. the optimal solutions to the utility maximization problem constrained to a budget set.
- $x^M(p_x, p_y, I)$  and  $y^M(p_x, p_y, I)$ .
- The indirect utility function is obtained by evaluating  $U(x,y)$  on the optimal solutions,

$$V(p_x, p_y, I) = U(x^M(p_x, p_y, I), y^M(p_x, p_y, I))$$

- The indirect utility function  $V(p_x, p_y, I)$  gives the maximum level of utility obtainable at prices  $p_x, p_y$  and income  $I$ .

# The indirect Utility function and the meaning of $\lambda$ .

- Let  $x^M$  and  $y^M$  denote the optimal values (demands) in the utility maximization problem. and let  $\lambda^*$  denote the optimal value of the multiplier
- Notice that we can write

$$\begin{aligned} V(p_x, p_y, I) &= U(x^M, y^M) = L(x^M, y^M, \lambda^*) \\ &= U(x^M, y^M) + \lambda^*[I - p_x x^M - p_y y^M] \end{aligned}$$

- We want to calculate  $\frac{\partial V}{\partial I}$ , i.e. how utility changes with respect to changes in income.

$$\begin{aligned} \frac{\partial V}{\partial I} &= U_x \frac{\partial x^M}{\partial I} + U_y \frac{\partial y^M}{\partial I} + \lambda^* \left( 1 - p_x \frac{\partial x^M}{\partial I} - p_y \frac{\partial y^M}{\partial I} \right) + (I - p_x x^M - p_y y^M) \frac{\partial \lambda^*}{\partial I} \\ &= \underbrace{(U_x - \lambda^* p_x)}_{=0 \text{ from the FOC's}} \frac{\partial x^M}{\partial I} + \underbrace{(U_y - \lambda^* p_y)}_{=0 \text{ from the FOC's}} \frac{\partial y^M}{\partial I} + \underbrace{(I - p_x x^M - p_y y^M)}_{=0 \text{ from the FOC's}} \frac{\partial \lambda^*}{\partial I} + \lambda^* \\ &= \lambda^* \end{aligned}$$

- Thus,  $\lambda^*$  is the marginal value, in terms of utility of a marginal increase in income.
- The result we just proved is a version of the Constrained Envelope Theorem (NS Ch. 2 pp. 45–46)